Predicting Wildfire Risk in California with Satellite Data

Abstract

Wildfires in California pose increasing threats to ecosystems, public health, and property, with climate change intensifying droughts and extending fire seasons. Predictive models are essential to anticipate fires before ignition and provide early warnings. This study evaluates the potential of satellite-derived vegetation indices (NDVI, EVI) and land surface temperature (LST) to predict wildfire occurrences during the 2020 California fire season. Using a curated dataset of over 18 million 1 km² samples with extreme class imbalance, we implemented Logistic Regression, Random Forest and Gradient Boosted Trees. Techniques such as SMOTE oversampling, class weighting, and probability threshold tuning were applied to address the rarity of fires. Gradient Boosted Trees achieved the highest recall, detecting over 90% of fire pixels, demonstrating that tree-based ensemble methods combined with data balancing can effectively identify rare wildfire events. Future work will integrate additional environmental variables and explore spatial-temporal modeling to enhance predictive performance.

Introduction

California wildfires are increasingly frequent and severe, threatening ecosystems, infrastructure and public health. Climate change has intensified droughts and extended fire seasons, making early detection and prediction critical for effective disaster management. Current satellite systems such as NASA's FIRMS provide near-real-time fire detection, but reactive monitoring is insufficient for prevention or early intervention.

This project aims to explore whether satellite-derived vegetation indices and land surface temperature can be used to predict wildfire occurrences before ignition. Vegetation health and moisture content measured through indices like NDVI and EVI are directly linked to fuel availability and dryness. Surface temperature, another key satellite variable, influences ignition likelihood by affecting fuel dryness and heat stress. By combining these features with historical fire data, we aim to train machine learning models capable of identifying high-risk zones, providing the foundation for proactive wildfire prediction.

Related Work

The first study below examines wildfire prediction within California's Central Valley using models such as Random Forest, Decision Trees, Naïve Bayes, and Neural Networks. Features include Air Temperature, Humidity, Precipitation and more. In contrast, the second study focuses on fire detection by leveraging data on fuel characteristics, ignition sources, and observed fire activity, aiming to reduce false-alarm rate and enhance fire forecast accuracy. Building on these approaches, our project uses complementary NASA datasets to explore how well vegetation and surface temperature data can be combined to predict wildfire risk.

[1] Applied ML with wildfire data in California's central valley - ScienceDirect
[2] Prediction of fire activity | Nature Communications

Dataset

To build a wildfire prediction model for California, we curated a multi-source satellite dataset covering the state during the 2020 lightning-induced wildfire outbreak, Jul 17 - Sept 18, chosen for its historical significance and completeness, in contrast to more recent fires that may lack finalized data. This dataset integrates three NASA Earthdata products:

The **Fire Detection Data** (FIRMS (Fire Information for Resource Management System)): active fire detections captured from the VIIRS sensor on the Suomi NPP satellite, with 375 meter spatial resolution. It includes the following attributes: fire location, brightness temperature, fire radiative power (FRP), confidence level, and acquisition time.

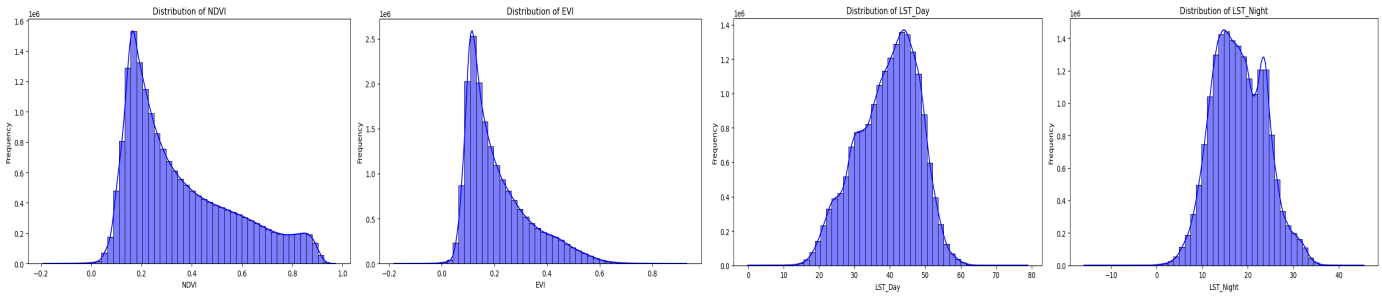
The **Vegetation Data** (MODIS MOD13Q1): 16-day composite vegetation indices from NASA's Terra satellite, specifically the Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI), with 250 meters of spatial

resolution. These indices are widely used to assess vegetation health, fuel availability, and dryness, factors that influence wildfire risk.

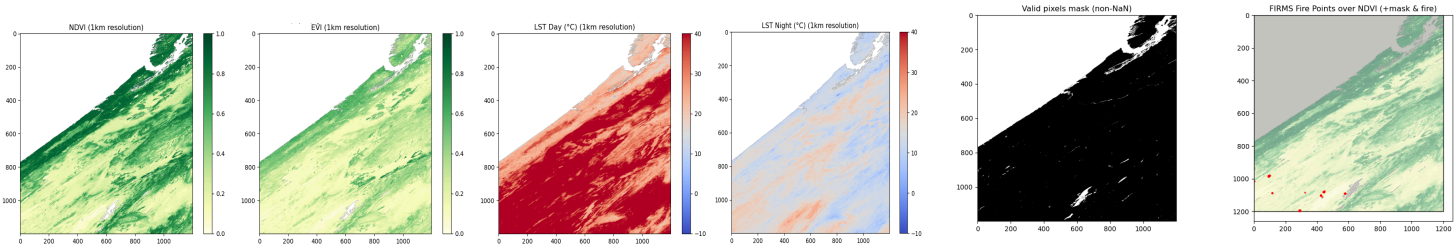
The **Surface Temperature Data** (MODIS MOD11A2): 8-day composite land surface temperature (LST) data at 1 kilometer spatial resolution. It includes both daytime and nighttime surface temperatures. Elevated LST can indicate heat stress, dryness, and favorable conditions for ignition and fire spread.

The data underwent extensive preprocessing to create a unified, clean dataset for modeling. We defined each sample as a 1-kilometer grid cell aligned with the MODIS system, encompassing four continuous input features: NDVI, EVI, LST Day, and LST Night. The binary target label—fire presence (1) or absence (0)—was generated by mapping FIRMS fire detection points onto the MODIS grid for each corresponding 16-day period. To handle mismatches in resolution and timing, the 250-meter vegetation data was downsampled to 1 kilometer by averaging, and two consecutive 8-day LST composites were averaged to match the 16-day vegetation windows. Invalid pixels, marked by fill values like -3000 for vegetation indices and 0 for LST, were identified, converted to NaN, and excluded using a comprehensive validity mask. This process, applied across five MODIS tiles covering California, yielded approximately 18.25 million valid samples from an initial 36 million, with only about 13,871 (0.08%) positive fire labels, highlighting the severe class imbalance. The final dataset was then randomly partitioned into training (70%), validation (15%), and test (15%) sets.

To understand patterns in the data, we conducted exploratory analysis on the four input features and their relationship to fire occurrence. Histograms of NDVI and EVI showed left-skewed distributions, with most values between 0.1 and 0.8, reflecting healthy vegetation across much of the landscape. Low values likely corresponded to deserts, burned or barren areas. LST Day values ranged broadly from 25-50°C, while LST Night showed a narrower distribution between 10-25°C, consistent with typical diurnal temperature patterns.

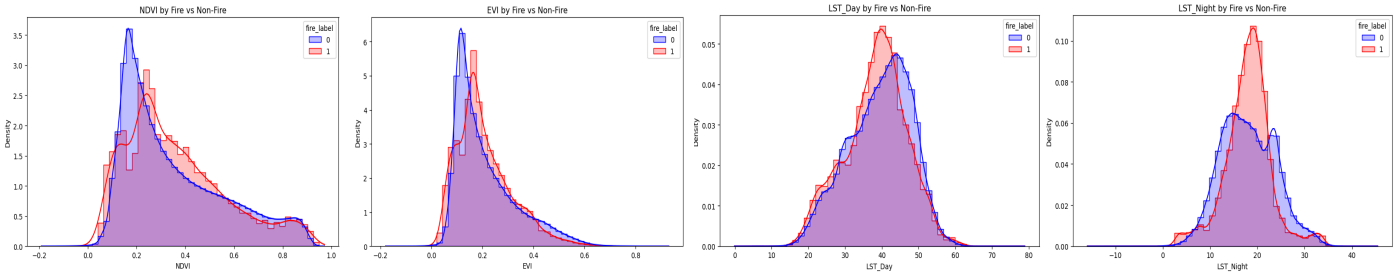


Spatial plots of feature layers revealed clear environmental gradients. For example, NDVI and EVI were higher in areas of dense vegetation (e.g., forests or cropland), while LST Day was elevated in arid or sparsely vegetated regions. These spatial patterns aligned with known ecological and climatic zones, supporting the physical validity of the satellite data.

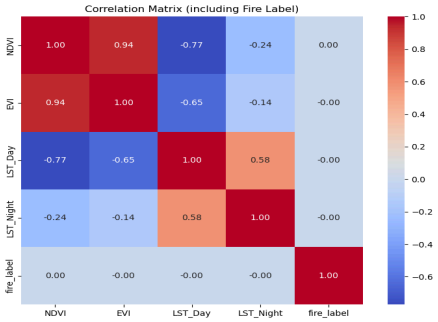


We visualized the fire mask, overlaying detected fire points from FIRMS onto the NDVI imagery along with transparency layers indicating both valid pixels and fire-affected regions. This confirmed that fire points were predominantly located in areas of lower vegetation density and good data coverage, supporting the spatial and physical reliability of our dataset.

We analyzed the distribution of each feature separately for fire and non-fire pixels using density histograms. Fire-labeled pixels are concentrated on lower NDVI/EVI values and higher LST Day temperatures, suggesting that fires are more likely in regions with sparse vegetation and elevated surface temperatures. These trends validate the physical relevance of our chosen features and indicate their potential predictive power for fire occurrence.



A correlation matrix showed strong positive correlation between NDVI and EVI (0.94) indicating redundancy in vegetation information, and strong negative correlations between NDVI and LST Day (-0.77), as well as EVI and LST Day (-0.65), due to the cooling effect of vegetation. LST Day and LST Night were moderately correlated (0.58). All input features had near-zero correlation with the binary fire label, suggesting that fire ignition depends on complex, nonlinear interactions.



Data Challenge

Several challenges arose in integrating these datasets into a single modeling framework. First, the temporal mismatch between MOD13Q1 (16-day composites) and MOD11A2 (8-day composites) required careful alignment. We resolved this by averaging pairs of LST layers to match the 16-day NDVI/EVI timeframes. Second, the spatial resolution mismatch between MOD13Q1 (250 m) and MOD11A2 (1 km) meant that vegetation indices had to be downsampled. We achieved this by computing the mean of 4×4 pixel blocks, aligning them with the 1 km LST grid. Lastly, incorporating FIRMS data required transforming geolocated fire points into the MODIS sinusoidal coordinate system. This process involved projection conversion, tile identification, and intra-tile pixel indexing. Only fire detections falling within the tile's spatial boundary and time window were used. We also enforced a validity mask to exclude any grid cells with missing or unreliable input data.

Methods

To establish a performance baseline, we first implemented **Logistic Regression (LR)** using TensorFlow. LR was chosen for its simplicity, interpretability, and efficiency in binary classification tasks such as predicting fire presence in a 1 km² grid cell. The model estimates the probability of fire occurrence as a linear combination of input features (NDVI, EVI, LST Day, and LST Night) and applies the logistic function to output probabilities between 0 and 1. To ensure proper scaling for this linear model, a Standard Scaler was applied to the training set features, preventing features with larger numeric ranges (e.g., LST) from dominating the learning process. Despite its simplicity, LR provides a useful benchmark to evaluate linear separability of fire vs. non-fire pixels.

Next, we implemented **Random Forest (RF)** using TensorFlow Decision Forests. RF is an ensemble method that builds multiple independent decision trees on bootstrap samples of the training data, combining their predictions through majority voting. Each tree selects features at random when splitting nodes, which increases model robustness and reduces variance. RF is appropriate for wildfire prediction because it captures nonlinear interactions between environmental variables, such as the combined effect of high daytime temperature and low NDVI on fire likelihood, which cannot be captured by linear models like LR. Importantly, tree-based models do not require feature scaling, so the Standard Scaler was not applied here.

Finally, we implemented **Gradient Boosting (GB)**, also using TensorFlow Decision Forests. Unlike Random Forest, GB builds trees sequentially, where each new tree focuses on correcting errors made by the previous ensemble. This allows the model to prioritize hard-to-predict fire pixels and capture subtle, nonlinear interactions, such as the interplay between

vegetation health indices and surface temperature anomalies. GB is particularly suitable for wildfire prediction due to its ability to handle imbalanced datasets and learn minority-class patterns during sequential tree-building.

All models were trained on the same dataset split:

- **Training:** 12,774,667 samples (9,710 fires)
- **Validation:** 2,737,429 samples (2,081 fires)
- **Test:** 2,737,429 samples (2,080 fires)

A critical challenge for all models was the extreme class imbalance, with fire pixels representing less than 0.01% of the total dataset. Any model could achieve high accuracy by predicting all pixels as “no fire.” To address this, we applied class weighting, SMOTE (Synthetic Minority Oversampling Technique), and threshold tuning as model-specific strategies to ensure meaningful learning of the minority fire class.

Experiments, Results and Discussion

For **Logistic Regression**, we addressed the severe class imbalance by first computing the class weights to assign higher penalties to misclassifications of the rare ‘fire’ class during training. We constructed a simple neural network with a single sigmoid-activated dense layer to perform binary classification, compiling it with binary crossentropy loss and Adam optimizer. The model was then trained for 10 epochs using these computed class weights. After training, we performed optimal threshold tuning by analyzing the precision-recall curve on validation set predictions to find the threshold that maximized the F1-score, moving above the default 0.5 threshold. These methods aimed to achieve a more balanced performance, maximizing recall where it could identify true fire events.

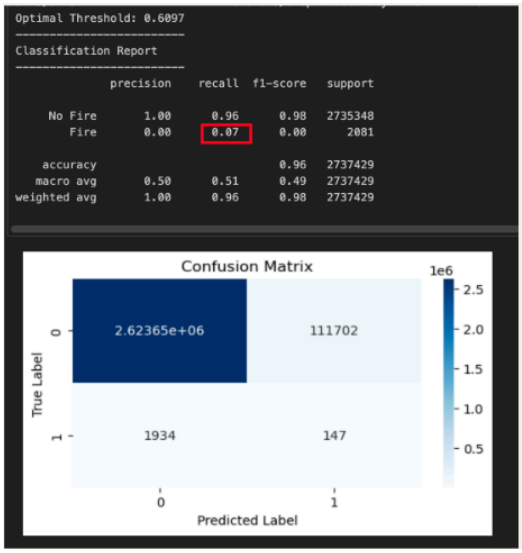
Next, we ran a **Random Forest** model where we addressed class imbalances by utilizing synthetic data generation techniques on the training data, specifically SMOTE (Synthetic Minority Over-sampling Technique) and ADASYN (Adaptive Synthetic Sampling). We imported the imblearn.over_sampling library to derive both SMOTE and ADASYN classes. The .resample method fitted our X_train_balanced and y_train_balanced dataframes. Then, we ran the RF model on all 3 datasets and compared the recall results for both fire and non-fire instances for all variations; Resulting in 58%, 86%, and 90% for correctly labelled fires, 86%, 60%, 55% for non-fire instances, and F-1 scores of 0.92, 0.75, and 0.71 (Train_balanced, SMOTE and ADASYN respectively).

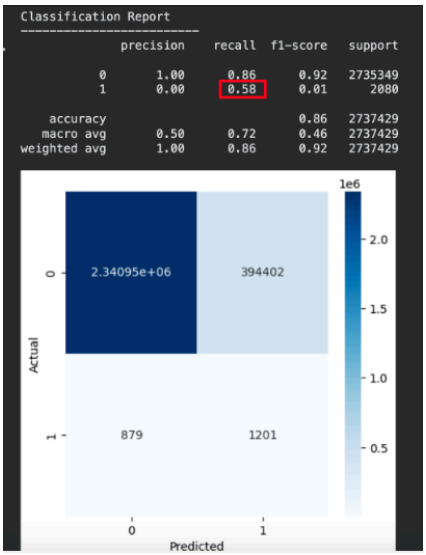
For **Gradient Boosting**, we applied SMOTE to the training set, increasing the fire class from 0.08% to 20% of the dataset, allowing the model to learn meaningful patterns from rare fire events. Class weights were also computed on the resampled dataset so fire pixels contributed more heavily to the loss function, reinforcing their importance during training. The GB model was trained with 500 trees of maximum depth 8, using binary cross-entropy loss, and a threshold of 0.3 was applied at inference to prioritize recall.

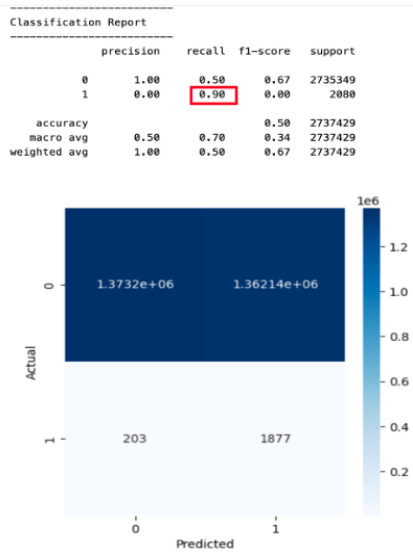
Evaluation was performed on separated validation and test sets, using metrics including recall, precision, F1-score, and confusion matrices, with performance analyzed separately for fire and non-fire classes. Gradient Boosting achieved ~90% recall for fire pixels, detecting nearly all fires, while precision remained near zero due to the high number of non-fire pixels misclassified as fire. Overall accuracy was ~50%, reflecting the trade-off between recall and false positives in extremely imbalanced datasets.

Overfitting was mitigated by limiting tree depth, using a large number of sequential trees, and combining SMOTE with class weights, which allowed the model to learn meaningful fire patterns without overfitting to noise in the minority class. Compared to Logistic Regression, which achieved only ~8% recall even with standard scaling, and Random Forest, which detected ~58% of fires, Gradient Boosting was the most effective model for our operational goal of early-warning wildfire prediction, where missing a fire is far costlier than generating false alarms.

All models were evaluated on the same validation set, which preserved the original severe imbalance. Performance was compared using precision, recall, F1-score, and confusion matrices. Overfitting was a primary concern given the dataset’s imbalance (<0.01% fire pixels), making it challenging for models to learn the patterns for minority class without mistaking noise for signal.







In our experiments, the Logistic Regression baseline with class weighting and optimized threshold correctly identified 147 true fires, but missed 1934 others, yielding a recall of approximately 8%. While this recall value appears low, it represents a significant result given the rarity of the event. This case also clearly illustrates the precision-recall trade-off, where precision approaches zero as recall increases. Followed by enhanced synthetic data strategies in the advanced models, Random Forest and Gradient Boosting were able to identify much higher fire counts (1201 and 1877, respectively) along with lower false fires (879 and 203, respectively).

Conclusion

This project developed and evaluated three machine learning models: Logistic Regression, Random Forest, and Gradient Boosted Trees for wildfire prediction across California using MODIS FIRMS fire data along with NDVI, EVI, and LST (day and night) features. Logistic Regression provided a baseline but was limited by its linear nature, detecting only a small fraction of fire events. Random Forest improved performance by capturing nonlinear relationships between environmental variables, detecting roughly 58% of fires. The Gradient Boosted Tree model, augmented with SMOTE oversampling, class weighting, and threshold tuning, achieved the best results, successfully identifying 90–91% of true fire pixels, making it the most effective approach for early-warning detection.

Despite high recall, precision remained low due to the extreme rarity of fire pixels. However, this trade-off is acceptable in operational contexts where missing a fire is far costlier than issuing false alarms. These results demonstrate that tree-based ensemble models, combined with data balancing techniques, are well-suited to predicting rare wildfire events from satellite imagery.

Several limitations constrain the current work. First, the extreme class imbalance, amplified by training across all of California, limits model precision and increases false positives. Second, computational constraints restricted experimentation with more sophisticated algorithms which could potentially improve performance. Third, the feature space was limited to four MODIS-derived variables; additional environmental factors such as wind speed, humidity, precipitation, topography, and drought indices would likely enhance the model's ability to distinguish fire vs. non-fire conditions.

Future work will focus on integrating additional NASA datasets that include meteorological and topographic variables, training models on specific high-risk regions with longer temporal coverage, and exploring more advanced machine learning approaches such as XGBoost, SVM, and spatial-temporal models (CNNs, RNNs, or Transformers). Probability thresholds can be further optimized using ROC and Precision-Recall curves to reduce false alarms while maintaining high recall. With these enhancements, wildfire prediction models could achieve improved accuracy and operational relevance, providing actionable early warnings to support disaster response and mitigation.

References

- [1] Hernandez, Kassandra, and Aaron B Hoskins. “Machine Learning Algorithms Applied to Wildfire Data in California’s Central Valley.” Trees, Forests and People, vol. 15, 1 Mar. 2024, pp. 100516–100516, https://doi.org/10.1016/j.tfp.2024.100516.

- [2] Giuseppe, Francesca Di, et al. “Global Data-Driven Prediction of Fire Activity.” Nature Communications, vol. 16, no. 1, 1 Apr. 2025, https://doi.org/10.1038/s41467-025-58097-7. Accessed 22 Apr. 2025.